

7 Kinematics	7.1	Understand and use the language of kinematics: position; displacement; distance travelled; velocity; speed; acceleration.	Students should know that distance and speed must be positive.
	7.2	Understand, use and interpret graphs in kinematics for motion in a straight line: displacement against time and interpretation of gradient; velocity against time and interpretation of gradient and area under the graph.	Graphical solutions to problems may be required.
	7.3	Understand, use and derive the formulae for constant acceleration for motion in a straight line. Extend to 2 dimensions using vectors.	Derivation may use knowledge of sections 7.2 and/or 7.4 Understand and use <i>suvat</i> formulae for constant acceleration in 2-D, e.g. $\mathbf{v} = \mathbf{u} + \mathbf{at}$, $\mathbf{r} = \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$ with vectors given in $\mathbf{i} - \mathbf{j}$ or column vector form. Use vectors to solve problems.
	7.4	Use calculus in kinematics for motion in a straight line: $\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$ $\mathbf{r} = \int \mathbf{v} \, dt, \quad \mathbf{v} = \int \mathbf{a} \, dt$ Extend to 2 dimensions using vectors.	The level of calculus required will be consistent with that in Sections 7 and 8 in the Pure Mathematics content. Differentiation and integration of a vector with respect to time. e.g. Given $\mathbf{r} = t^2\mathbf{i} + t^{\frac{3}{2}}\mathbf{j}$, find $\dot{\mathbf{r}}$ and $\ddot{\mathbf{r}}$ at a given time.
	7.5	Model motion under gravity in a vertical plane using vectors; projectiles.	Derivation of formulae for time of flight, range and greatest height and the derivation of the equation of the path of a projectile may be required.